Project Title: Smart Electricity Consumption Prediction & Energy Optimization System

Task ID: ML-3

Domain: Machine Learning

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1. Dataset Creation & Summary

In strict compliance with Teyzix Core's policy of not using public datasets, an original dataset of 1500 records was programmatically generated using the Python `Faker` library. The dataset simulates real-world electricity consumption patterns, specifically modeling intense summer weather conditions (similar to regions like Wah Cantt) and varying appliance usage. It includes 15 realistic features such as House Type, Family Members, AC Usage, Fan Usage, Outdoor Temperature, and an explicitly calculated target variable: `Daily Consumption (kWh)`.

2. Data Preprocessing & Exploratory Data Analysis (EDA)

The raw dataset was cleaned and optimized for machine learning. Missing values were handled using median imputation, and categorical variables (House Type, Season, Day of Week) were processed using One-Hot Encoding. Numeric features were standardized using `StandardScaler`. During EDA, the correlation heatmap revealed that `AC Usage (Hours)` and `Outdoor Temperature (C)` had the most significant positive correlation with the daily electricity consumption.

3. Model Training & Performance Evaluation

Three advanced regression algorithms were trained and compared: Linear Regression, Random Forest Regressor, and XGBoost.

- **Performance Insight:** Linear Regression achieved the highest R^2 score (near perfect), closely followed by Random Forest (97%). The exceptional performance of Linear Regression is attributed to the direct linear mathematical relationship formulated during the synthetic dataset generation (Consumption = Appliance 1 + Appliance 2, etc.).
- The system dynamically selected the best-performing model, preserving its state and the `StandardScaler` using `joblib` for live deployment.

4. Prediction System & Energy Forecasting

An interactive web application was deployed using **Gradio**. The system takes real-time household and appliance data as input, dynamically applies the `.reindex()` method to prevent dimensional mismatch, and predicts:

- Predicted Daily Consumption
- Estimated Monthly Usage and Electricity Bill (Calculated at ~45 PKR/kWh)

- Energy Efficiency Score (Per Capita Usage)

5. Energy Optimization Recommendations

To provide actionable business intelligence, the system acts as a smart consultant. Based on the real-time inputs, it generates personalized recommendations such as shifting heavy appliance loads (washing machines, water motors) outside of peak hours, optimizing AC temperatures to 26°C, and upgrading to energy-efficient lighting.